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PALAEOMAGNETISM OF THE DYKE SWARMS OF THE GARDAR IGNEOUS PROVINCE, SOUTH GREENLAND

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In the western part of the Gardar Igneous Province of southern Greenland, lamprophyre dykes intruded at ca. 1276–1254 m.y. (Rb–Sr biotite ages) yield a palaeomagnetic pole at 206.5°E,3°N (nine sites, $d\Psi = 5.1°$, $d\chi = 10.1°$). Slightly younger dolerite dykes with Rb–Sr biotite ages in the range 1278–1263 m.y. give a pole at 201.5°E,8.5°N (24 sites, $d\Psi = 4.7°$, $d\chi = 9.4°$), and the syeno-gabbro ring dyke of the Kûngnât complex (Rb–Sr isochron age 1245 ± 17 m.y.) cutting both of these dykes swarms, gives a pole at 198.5°E,3.5°N (four sites, $d\Psi = 2.3°$, $d\chi = 4.4°$). All these rock units have the same polarity and the poles are identical to those from Mackenzie and related igneous rocks of North America (1280–1220 m.y.) after closure of the Davis Strait; they confirm that this part of the Gardar Province is a lateral extension of the Mackenzie igneous episode within the Laurentian craton.

In the Tugtutôq region of the eastern part of the Gardar Province 47 NNE-trending dykes of various petrologic types, and intruded between 1175 ± 9 and 1168 ± 37 m.y. (Rb–Sr isochron ages) yield a palaeomagnetic pole at 223.9°E,36.4°N ($d\Psi = 4.1°$, $d\chi = 6.1°$). Fifteen other dykes in this swarm were intruded during a transitional phase of the magnetic field which, however, does not appear to have achieved a complete reversal over a period of several millions of years. The majority of dykes studied are highly stable to AF and thermal demagnetisation and contain single high blocking temperature components with single Curie points in the range 380–560°C.

Palaeomagnetic poles from the Gardar Province between ca. 1330 and 1160 m.y. in age define the earlier part of the Great Logan apparent polar-wander loop; they correlate closely with contemporaneous North American results and confirm the coherence of the Laurentian craton in Upper Proterozoic times.

1. Introduction

Upper Proterozoic time (ca. 1300–600 m.y.) was characterised by widespread anorthositic, alkaline and related magmatism within the continents bordering the present North Atlantic Ocean, and palaeomagnetic studies of these rocks have yielded considerable information on apparent polar movements and geomagnetic field behaviour during this interval (e.g., Irving and Lapointe, 1975). Many uncertainties remain, however, in the timing and direction of apparent polar movements primarily because igneous events within any one area were concentrated within relatively short periods of time. In this paper we extend the study of Upper Proterozoic palaeomagnetism to Greenland where detailed studies of the Gardar Igneous Province yield results which help to clarify some of these problems during the intervals ca. 1330–1245 and ca. 1200–1160 m.y.

2. Dyke swarms of the western part of the province

The Gardar igneous cycle was originally defined by Wegmann (1938) to incorporate the widespread Upper Proterozoic magmatic activity in Greenland south of latitude 61.5°N. The cycle commenced with eruption of basaltic lavas and formation of continental sandstones prior to 1310 m.y. and was followed by intrusion of a number of peralkaline bodies within the interval 1330–1160 m.y. The other important facet of the Gardar cycle was the emplacement of swarms of dykes

along E–W and ESE–WNW, and later NE trends.

In this section we cover the palaeomagnetism of Gardar rocks from the Arsûk–Ivigtut area NW of Kobberminebugt, a region in which igneous activity took place appreciably earlier than in the eastern part of the province (van Breemen and Upton, 1972; Blaxland et al., 1977). The Gardar rocks here intrude a basement complex of Archaean gneisses and rocks of the Ketilidian mobile belt (ca. 1800 m.y.). The oldest igneous episode is the Grønnedal–Ika carbonatite complex dated 1327 ± 17 m.y. by Blaxland et al. (1977) which is cut by some members of a regional dolerite dyke swarm. These dykes have trends between WNW and WSW, and progressive reorientation of the regional stress field took place during intrusion so that WNW members are cut by E–W- and WSW-trending members. They were sampled for palaeomagnetic studies at 48 localities (Fig. 1) and cut some members of a swarm of thin and impersistent lamprophyre dykes which also postdate the Grønnedal–Ika complex; the lamprophyres were sampled for palaeomagnetic study at fourteen sites. Patchett and Upton (1977) have determined Rb–Sr biotite ages of 1276 and 1254 m.y. for the lamprophyre dykes and 1278, 1277, 1272, 1265 and 1263 m.y. for the dolerite dykes.

Both these swarms are in turn cut by the Kûngnât Fjeld syenite complex (Fig. 1) dated 1240 ± 150 m.y. (Rb–Sr age on biotite) by Moorbath et al. (1960) and 1245 ± 16 m.y. (Rb–Sr whole-rock isochron) by van Breemen and Upton (1972); this age was recalculated to 1233 m.y. ($\lambda = 1.39 \cdot 10^{-11}\ a^{-1}$) by Blaxland et al. (1977) who determined a subsequent Rb–Sr isochron of 1245 ± 17 m.y. on this body. Late faulting of this complex permitted intrusion of a syeno-gabbro ring dyke approximately 100 m wide and sampled at four localities for this study across the width of the dyke. Intrusion of the whole complex is known to have taken place within a short period of time since late fractionates from the syenite are seen to intrude the ring dyke (Upton, 1960).

The relationship of these rocks to the other Gardar intrusion of this area, the Ivigtut granite from which Moorbath and Pauly (1962) derived a whole-rock Rb–Sr age of 1330 ± 20 m.y. and A.B. Blaxland (personal communication, 1976) has subsequently obtained an Rb–Sr isochron of 1248 ± 25 m.y., is not completely clear. This granite is, however, cut by a number of small ENE-trending dykes (Berthelesen, 1960) and appears to predate at least the later members of the regional dyke swarm.

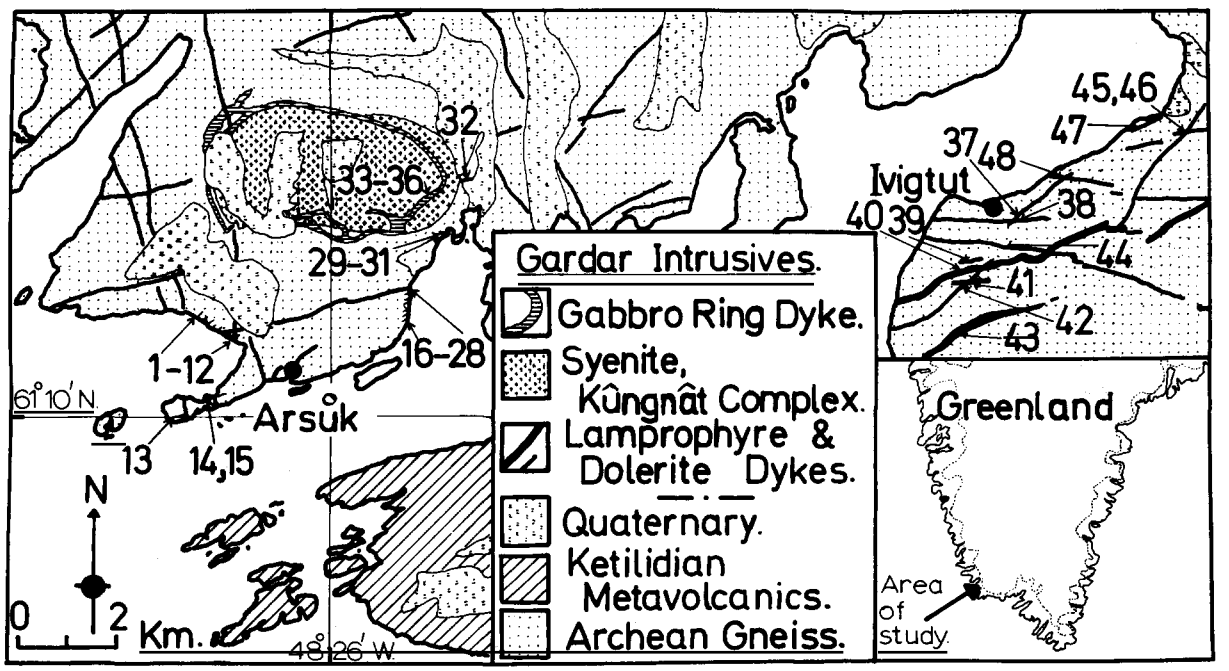


Fig. 1. Sampling sites for this study in the western part of the Gardar Igneous Province, southwest Greenland.

The samples for this study were drilled in the field using a motorised portable drill (Doell and Cox, 1967) and oriented while still attached to the outcrop by sunsights and/or magnetic bearings in conjunction with topographic sights. The rock cores were subsequently sliced into cylinders 2.5 cm in length and their magnetisations measured with a parastatic magnetometer system.

3. Palaeomagnetic results from the western region

Each core was stepwise demagnetised in AF's in steps of 50–100 Oe to maximum peak fields between 500 and 1,400 Oe until no further systematic change in magnitude or direction of natural remanent magnetisation (NRM) was observed. Of the lamprophyre dykes (Fig. 2, Table I) samples with strong total NRM's typically exhibit little or no change in direction with progressive treatment; other samples with initial directions near the present geomagnetic field move to shallow westerly directions with progressive treatment, and finally some samples with both variable total NRM's and magnetic stabilities move to appreciably different directions and stabilise with directions remote from most other samples (Fig. 3). These sites were not used for computing the group mean, although since

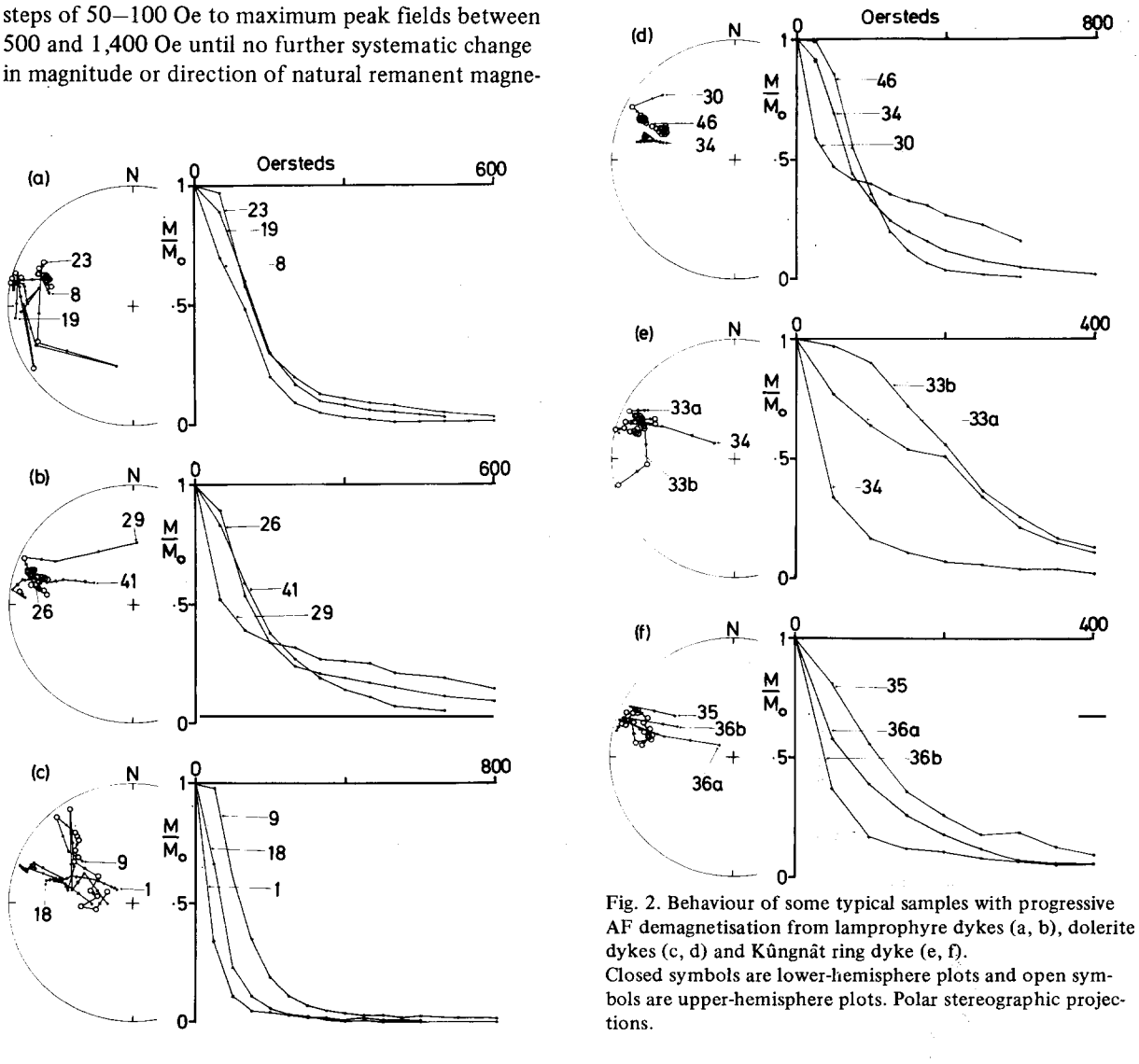


Fig. 2. Behaviour of some typical samples with progressive AF demagnetisation from lamprophyre dykes (a, b), dolerite dykes (c, d) and Kûngnât ring dyke (e, f).
Closed symbols are lower-hemisphere plots and open symbols are upper-hemisphere plots. Polar stereographic projections.

TABLE I

Palaeomagnetic results from the western part of the Gardar Igneous Province, south Greenland (311.7°E,61.2°N)

Site	*N*	Cleaning field	*I* (°)	*D* (°)	*R*	*k*	α_{95} (°)	M_0 (10^{-4} e.m.u./g)	χ ($\times 10^{-3}$)
(i) Lamprophyre dykes									
2 *	2	250	41.8	279.5	1.97	–	–	5.0–10.3	2.8–4.1
8	6	250	−0.1	279.4	5.95	94	7.0	6.9–8.5	2.1–2.3
12 *	6	150	16.2	71.1	5.68	15	17.6	2.6–4.2	2.0–2.7
13 *	6	200–250	51.2	293.5	5.92	64	8.5	7.0–13.9	2.3–2.4
15 *	6	400	48.9	291.7	5.91	53	9.3	0.6–1.2	0.3–1.2
16	6	150	17.3	283.7	5.88	42	10.4	5.9–18.0	3.6–4.9
17 *	6	150	75.5	21.1	5.83	29	12.6	17.2–37.2	4.0–5.9
19	6	150	1.7	284.8	5.97	147	5.6	1.3–2.5	1.2–1.4
23	6	200	−23.1	286.8	5.98	203	4.7	5.9–32.9	1.7–5.3
25	5	150–200	−24.5	288.8	5.00	107	2.3	10.8–24.4	1.7–2.4
26	6	150	−21.4	287.0	5.99	895	2.2	6.4–18.7	1.1–1.9
29	6	300	−7.2	288.9	5.88	40	10.7	0.4–2.5	0.3–2.6
31	6	200	−22.7	286.7	5.95	103	6.7	3.7–32.3	2.6–3.2
41	4	250–500	3.2	278.3	3.97	90	9.4	1.7–4.2	1.0–1.3
(ii) Dolerite dykes									
1	6	250	32.3	274.8	5.87	38	11.0	2.6–6.2	7.4–8.6
3	6	150	−7.4	303.0	5.89	44	10.2	3.9–7.1	1.3–2.2
4	6	200	−33.2	288.9	5.94	78	7.6	3.2–3.7	1.3–1.9
5 *	6	150	−33.5	79.9	5.97	159	5.3	2.8–10.8	2.0–2.3
6	6	200	−39.2	292.5	5.92	64	8.5	2.1–3.2	1.0–1.2
7	6	200	21.6	300.8	5.88	43	10.4	2.2–3.4	0.7–0.8
9	6	200–250	−21.1	316.4	5.95	105	6.6	3.6–7.7	1.6–1.9
10	6	150–200	41.2	290.0	5.91	55	9.1	5.2–18.9	1.3–1.9
11	5	150	11.7	299.2	4.98	247	4.9	1.3–2.5	0.8–1.1
14 *	5	200	20.7	74.4	4.93	58	10.1	4.0–7.9	3.3–4.4
18	6	300	11.9	281.5	5.98	241	4.3	1.9–8.0	1.4–3.1
20 *	6	100	66.1	277.4	5.93	75	7.8	0.02–0.4	0.02–0.3
21 *	5	150	66.8	323.4	4.97	119	7.0	22.5–95.5	0.2–1.2
22	5	150–200	11.8	292.7	4.97	146	6.4	0.4–2.3	1.4–16.0
24	6	100	−26.1	287.3	5.99	892	2.2	10.1–16.0	1.7–2.3
27	5	200	−19.4	290.7	4.98	184	5.7	3.9–12.4	1.6–1.9
28	6	300	−28.4	288.2	5.99	588	2.8	27.2–46.1	0.7–0.9
30	6	300	−16.6	296.8	5.99	530	2.9	1.3–2.5	0.6–0.9
32	6	400	−28.7	300.4	5.97	175	5.1	3.4–4.4	1.0–1.0
37	8	300	1.1	305.3	7.66	21	12.5	0.3–3.5	0.1–1.0
38	5	300–350	−1.6	296.1	4.91	44	11.7	1.5–8.5	0.3–1.4
39	6	300	21.1	285.6	6.00	2,144	1.5	15.2–21.7	0.7–1.3
40	5	250	14.2	281.6	4.99	561	3.2	5.5–6.8	0.6–0.7
42	6	200–300	−7.3	288.4	5.89	46	10.0	5.2–7.6	1.3–1.5
43	6	300	28.0	296.1	5.99	380	3.4	2.4–3.1	0.3–0.4
44 *	3	50–100	30.8	236.6	2.97	65	14.4	0.04–0.02	0.02–0.03
45	5	200	−13.6	275.8	5.00	1,029	2.4	5.9–7.7	1.3–1.6
46	5	250–300	−30.5	283.9	4.99	359	4.0	12.6–15.8	15.0–2.1
47 *	6	300	86.2	314.8	5.99	347	3.6	7.2–9.5	1.0–1.4
48	5	200	4.5	278.2	4.87	32	13.8	1.8–15.1	0.8–2.5
(iii) Kûngnât ring dyke									
33	5	400	−13.3	290.4	4.99	729	2.8	6.0–12.8	1.0–1.9
34	5	200–250	−19.7	295.3	4.86	28	14.8	0.3–5.1	0.02–1.3
35	3	300–350	−15.1	292.5	2.97	65	15.4	1.5–15.9	0.06–1.5
36	5	200–250	−13.6	289.8	4.97	154	6.2	1.4–5.9	0.6–1.8

TABLE I (*continued*)

	N	I (°)	D (°)	R	k	α_{95} (°)	Lat. (° (°N)	Long. (°E)	$d\Psi$	$d\chi$
Group mean directions and poles										
(i) Lamprophyre dykes	9	−8.6	284.8	8.71	28	10.0	3.2	206.4	5.1	10.1
(ii) Dolerite dykes	24	−3.3	291.5	21.88	11	9.4	8.7	201.7	4.7	9.4
(iii) Kûngnât ring dyke	4	−15.4	292.0	3.99	455	4.3	3.4	198.7	2.3	4.4

R = resultant vector; D = declination and I = inclination of the mean direction derived from N samples (or sites); k = precision estimate [= $(N-1)/(N-R)$]; α_{95} = radius of the 95% confidence circle about the mean direction; M_0 = total natural remanent magnetisation (NRM); and χ = mass susceptibility in c.g.s. units.
* Sites excluded from the group mean calculation.

two of them are very stable we believe that they record intermediate directions of the palaeofield. Sites from the dolerite dykes fall into the same three divisions, with three sites exhibiting good stability which yield intermediate directions remote from the main group. Most of these samples stabilise with 10% or less of initial moments remaining (Fig. 2) and further thermal treatment of these residual moments exhibits no further systematic changes in remanence direction (Fig. 4) although in general samples used in this study were less stable to thermal than to AF treatment, possibly due to alteration of primary sulphides promoted by heating; shoulders in the demagnetisation spectra indicate single high temperature blocking temperature components in the range 450–550°C.

Although a change in orientation of the regional stress field took place during intrusion of the dolerite dyke swarm, the older dykes with WNW orientations and cut by later E–W- and WSW-trending dykes (Fig. 1) do not have significantly different directions of magnetisation (cf. Fig. 1 and Table I) and we infer that the change in regional stress field took place in an interval

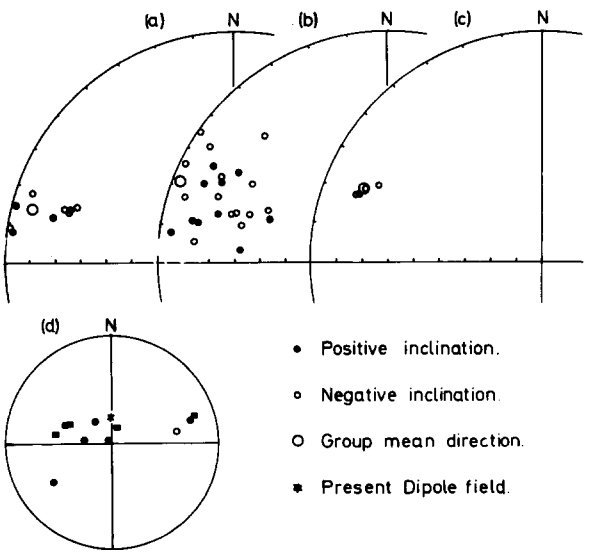


Fig. 3. Site mean directions of magnetisation of: (a) lamprophyre dykes; (b) dolerite dykes; and (c) Kûngnât ring dyke. The site mean directions of magnetisation not included in the group mean directions are plotted in (d).

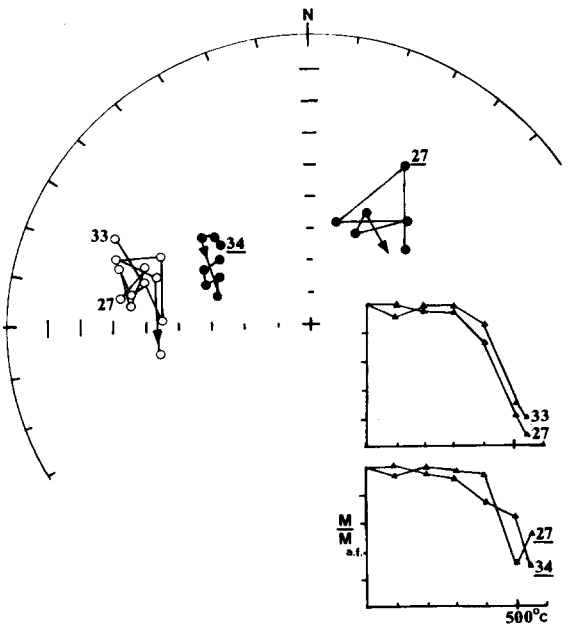


Fig. 4. Some examples of the thermal demagnetisation of residual moments in pilot samples together with demagnetisation curves. Underlined numbers are from the eastern part of the Gardar Province.

during which the predominant magnetic field was stationary. The age data outlined in Section 2 show that this interval may have lasted for an interval up to ~30 m.y.; contemporaneous North American data (Irving and Hastie, 1975) also suggest that this time period was an interval of constant polarity and pole position (Section 6).

The four sites from the Kûngnât ring dyke yield statistically significant directions of magnetisation with good grouping after AF cleaning, with a group mean direction similar to those from the lamprophyre and dolerite dykes. Site and group mean directions after cleaning are listed in Table I and the lamprophyre, dolerite and Kûngnât poles are not significantly different at the 95%-confidence level. With the exception of the intermediate directions all rock units are of constant polarity. The site mean statistics listed in Table I are computed after cleaning all samples in a field selected from the behaviour of one or more pilots from each site; the cleaning field was selected as that field beyond which no further systematic or significant change in remanence direction was observed.

4. Dyke swarms of the eastern part of the province

In the eastern part of the Gardar Igneous Province only a few intrusions are of comparable age to the rocks in the west of the province discussed in Sections 2 and 3. Most episodes are 100–150 m.y. younger (Blaxland et al., 1977), and intrusions with tectonic lineaments are oriented along NW–SW trends by way of contrast to the WNW–WSW trends of dykes in the western part of the province (Fig. 5). The area under consideration here lies between 46–47°W and 60–61°N, and is confined to the island of Tugtutôq. The WNW–ESE-trending dykes of the western part of the province are represented in this region by a few isolated dolerite dykes (denoted in the geological litera-

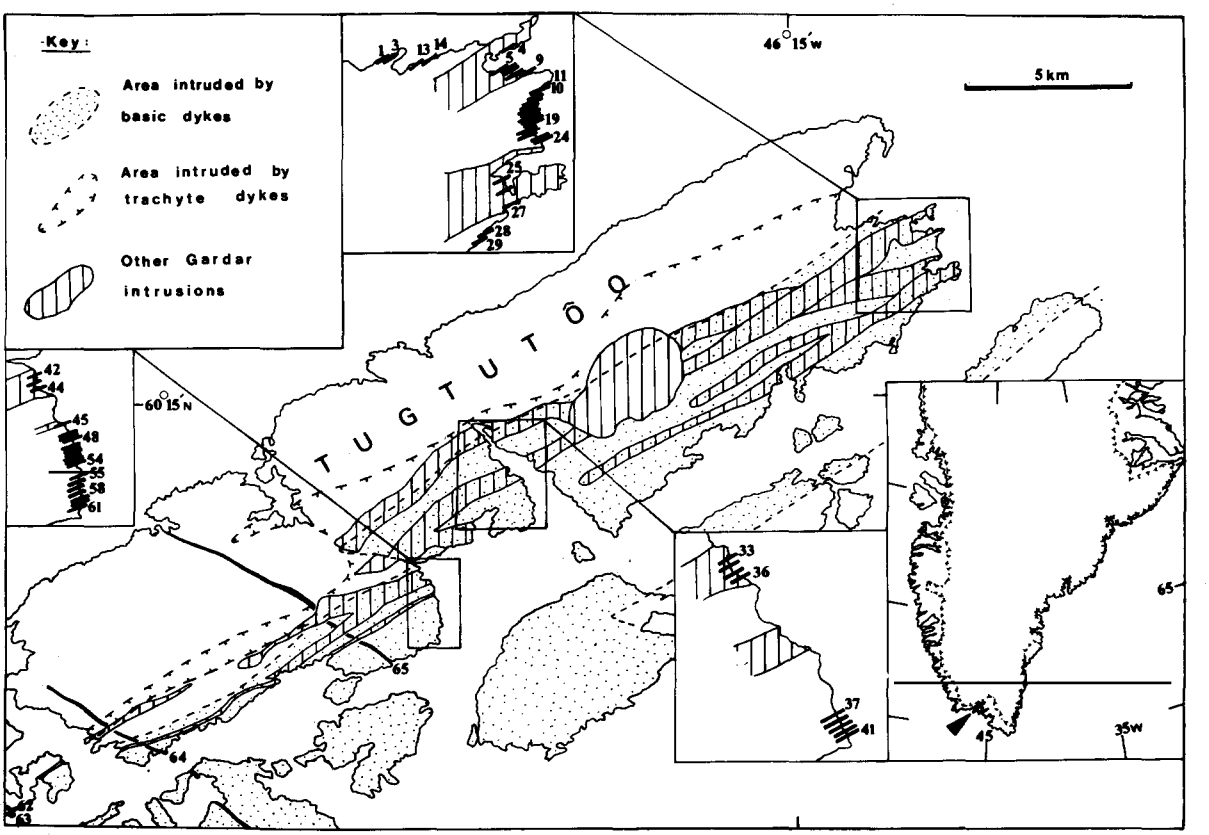


Fig. 5. Sampling sites in dyke swarms of the Tugtutôq region of the Gardar Igneous Province, south Greenland.

ture as BDO dykes) sampled for palaeomagnetic study at three localities (Fig. 5). Patchett and Upton (1977) determined Rb–Sr isochrons of 1285 ± 144 and 1244 ± 117 m.y. on two of these dykes confirming that they are similar in age to dykes of comparable trend in the western part of the province. The earliest intrusion with a NE–SW trend is the Hviddal giant dyke dated 1187 ± 9 m.y. by van Breemen and Upton (1972) and recalculated to 1175 m.y. by Blaxland et al. (1977) using revised constants for the analytical procedure. Intrusion of this dyke was closely followed by the intrusion of several gabbro giant dykes which are in turn cut by numerous NE–SW dykes of varying composition and which form the subject of this paper. All these rocks predate the Tugtutôq central complex dated by Rb–Sr whole-rock isochron at 1180 ± 37 m.y. by van Breemen and Upton (1972) and recalculated to 1168 m.y. by Blaxland et al. (1977). This complex represents the final episode of Gardar magmatism in the Tugtutôq region although intrusion of alkaline complexes continued until ca. 1160 m.y. on the mainland at Ilímaussaq and Klokken (Blaxland and Parsons, 1975; Blaxland et al., 1976).

The NE–SW-trending dykes are exposed over a zone approximately 8 km wide and are concentrated along the SE part of the island of Tugtutôq (Fig. 5) and the Ilímaussaq peninsula. Sixty-two dykes were sampled in coastal sections cutting this swarm as indicated in the inset diagrams of Fig. 5. The dykes occur as swarms of varying petrology including trachytes, trachydolerites, dolerites, saturated and undersaturated microsyenites, and alkali microgranites; some are aphyric while others, including members described as "big-feldspar dykes", contain plagioclase and sometimes anorthosite xenocrysts (Upton, 1964). Palaeomagnetic sampling was concentrated on the more basic members (Table II). Intrusion of these dykes evidently spanned a considerable period of time since earlier members are sometimes seen to be cut by later members, although the total period of intrusion may not be more than a few millions of years since most, if not all, dykes postdate the Hviddal giant dyke (1175 m.y.) and predate the Tugtutôq central complex (1168 m.y.).

5. Palaeomagnetic results from the eastern region

All samples were measured before and after stepwise treatment in AF's in steps of 50 Oe up to peak fields between 400 and 1,000 Oe, and the behaviour of magnetic moments and directions of magnetisation with this treatment for three samples from each of eight typical dykes is illustrated in Fig. 6. With low-field treatment the directions typically move to shallower inclinations and stabilise in fields ranging 100–800 Oe with little or no further change in direction. At this stage 1–10% of the total NRM remains in most cases (Fig. 6) and to test the stability of these residual moments pilot samples were again subjected to progressive thermal demagnetisation. No further systematic change in direction is evident and the demagnetisation curves again demonstrate (Fig. 4) that single high blocking temperature components in titanomagnetite are responsible for the remanence in these rocks. After cleaning, all dykes exhibit good grouping of directions significant at the 95%-confidence level and are listed in Table II; sample directions here have been averaged over the most stable range of behaviour during demagnetisation.

The dyke mean directions of magnetisation are predominantly of moderate to steep positive inclination and WNW declination (Fig. 7), and the 47 sites falling in this well-defined group yield a group mean direction of $D = 289.6°$, $I = 50.6°$ ($\alpha_{95} = 4.5°$) with a derived palaeomagnetic pole at 223.9°E,36.4°N. No dykes have directions of magnetisation antiparallel to this group and it is inferred that they were intruded during a period of constant geomagnetic field polarity at ca. 1175–1168 m.y. Fifteen of the dykes do, however, have group mean directions of magnetisation removed by 30° or more from the main group of 47 sites, and since within-site groupings are in most cases of comparable quality to the remaining dykes (Table II), it is concluded that they record intermediate directions of the geomagnetic field. These dykes have predominantly ENE declinations and shallow to steep inclinations (Fig. 7), and include members of all the petrologic divisions (Table II). The nature of these intermediate directions in the context of geomagnetic field behaviour is not completely clear since only one polarity is evident from the bulk of the collection; although intrusion of the dykes occupied a considerable period of time, no reversals have been found. We have the unusual situation where transitional field directions are well represented and reversed field directions are unrepresented. In view of the large size of the sample, this seems unlikely to have originated by chance, and

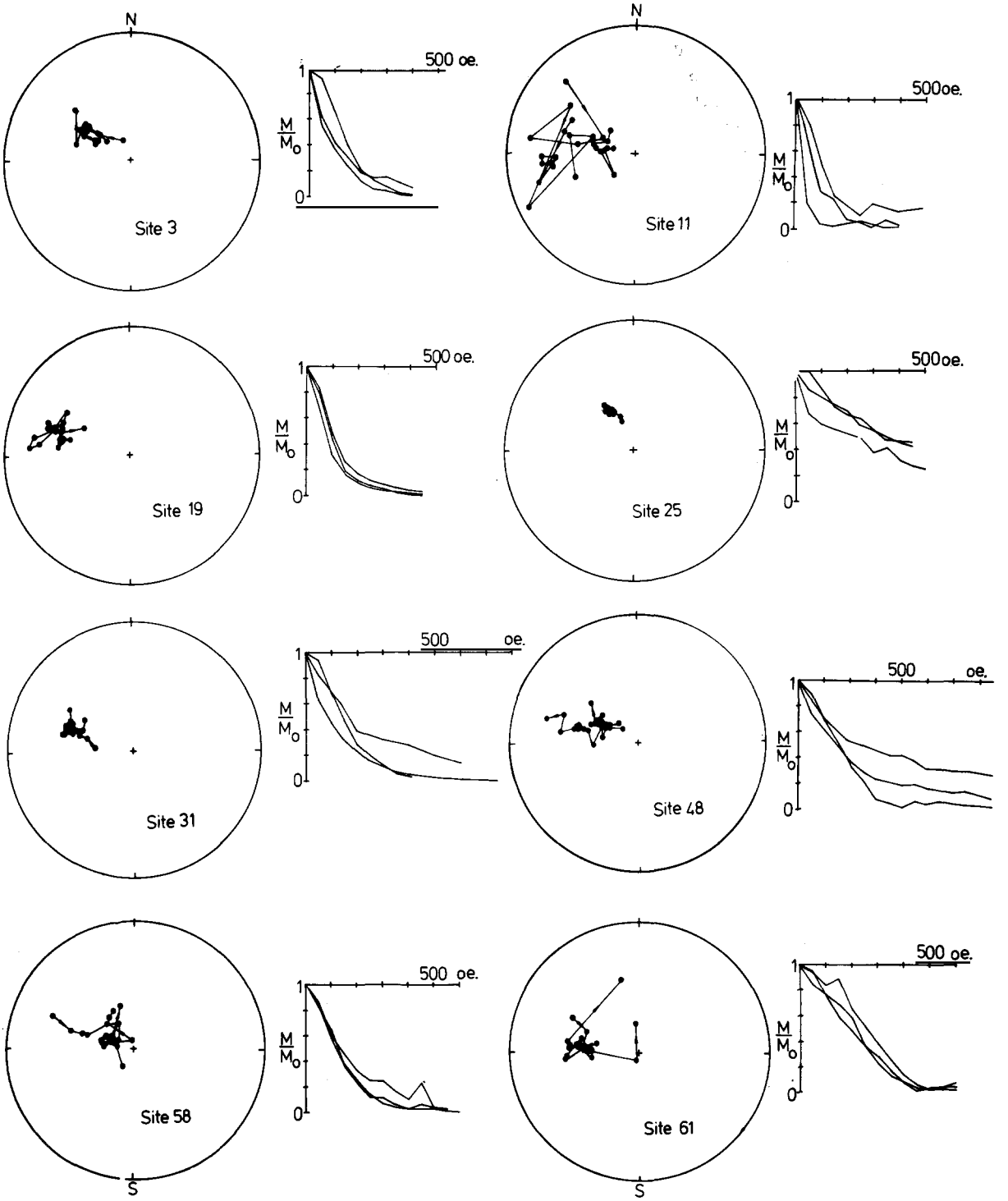


Fig. 6. Behaviour of directions of magnetisation and magnetic moments for three samples from each of eight sites in NE-trending dykes.

TABLE II

Palaeomagnetic results, dyke swarms of eastern part of Gardar Igneous Province, south Greenland (313.7°E,60.8°N)

Site	Type of dyke	N	D (°)	I (°)	R	k	α_{95} (°)	M_0 (10^{-5} e.m.u./g)
(i) NE–SW dykes								
1	ABD	5	302.8	41.7	4.89	37	12.7	2.86–4.88
2	FPD	6	315.4	40.5	5.66	15	18.0	0.97–1.24
3	ABD	4	307.6	50.1	3.95	59	12.1	1.85–6.77
4	TRD	6	308.8	21.2	5.18	6	29.6	0.01–0.03
5	TRD	5	270.4	72.7	4.79	19	18.0	0.02–0.20
6	TRD	2	159.3 *1	65.4	1.98	–	–	0.02–0.26
7	TRD	3	309.1	51.8	2.91	11	39.9	0.05–0.17
8	TRD	5	100.5 *1	64.9	4.81	21	16.9	0.05–0.67
9	ABDX	5	297.7	60.6	4.94	62	9.8	0.02–0.06
10	BFD *2	6	298.9	56.5	5.91	53	9.3	2.54–3.80
11	ABD	6	272.8	45.9	5.74	19	15.7	3.00–6.50
12	ABD	4	271.4	39.9	3.92	39	14.8	4.03–21.87
13	ABD	5	266.6	43.3	4.85	26	15.1	3.27–36.90
14	ABD	6	274.3	38.8	5.67	15	17.7	0.78–3.09
15	ABD	6	272.7	45.6	5.98	244	4.3	0.51–12.21
16	ABD	6	275.6	39.7	5.97	197	4.8	4.70–11.63
17	ABD	5	290.4	48.8	4.91	44	11.6	5.30–12.34
18	ABD	7	286.9	50.9	6.85	41	9.5	6.79–11.40
19	TRD	6	283.7	35.7	5.98	254	4.2	4.89–20.34
20	ABD	6	306.8	30.9	5.95	94	6.9	0.51–1.12
21	ABD	5	160.0 *1	72.2	4.65	12	23.5	0.03–0.51
22	ABD	6	45.3 *1	44.2	5.82	27	13.1	0.51–1.60
23	ABD	5	266.2	39.6	4.65	11	23.8	2.85–12.73
24	ABD	6	278.1	19.2	5.82	28	12.9	2.19–5.73
25	BFD	6	331.8	59.7	5.99	407	3.3	0.06–0.33
26	BFD	5	83.4 *1	4.3	4.95	86	8.3	0.03–0.08
27	FPD	6	30.3 *1	42.8	5.95	107	6.5	0.05–0.28
28	TRD	6	69.5 *1	34.1	5.96	128	5.9	0.60–2.48
29	ABD	6	289.2	48.8	5.84	31	12.2	0.18–1.62
30	BFD	6	309.3	63.1	5.96	121	6.1	2.31–9.57
31	ABD	6	282.9	47.8	5.94	87	7.2	7.31–10.38
32	TRD	6	71.5 *1	49.6	5.74	19	15.7	0.01–0.02
33	FPD	6	311.0	60.2	5.77	21	14.8	0.01–0.02
34	BFD	5	317.0	46.6	4.84	25	15.5	0.14–4.73
35	BFD	6	301.7	47.0	5.57	12	20.5	0.01–0.35
36	BFD	4	284.6	40.9	3.98	124	8.3	0.61–5.31
37	ABD	5	70.8 *1	51.0	3.95	4	45.4	0.09–1.30
38	ABD	5	63.3 *1	13.8	4.80	20	17.6	0.77–8.21
39	FPD	6	290.1	33.5	5.72	18	16.2	1.36–6.67
40	ABD	6	277.5	33.6	5.94	90	7.1	4.31–6.08
41	BFD	7	287.8	33.7	6.81	32	10.2	0.48–2.17
42	FPD	6	299.2	71.3	5.92	65	8.4	0.07–0.47
43	ABD	6	319.8	58.1	5.84	31	12.2	0.89–5.59
44	ABD	7	273.4	71.5	6.86	43	9.3	0.05–0.49
45	ABD	6	281.2	62.5	5.94	80	7.6	0.07–0.59
46	ABD	6	87.2 *1	–3.0	5.69	16	17.1	0.58–1.26
47	TRD	5	85.1 *1	–9.2	4.67	12	23.0	1.20–9.01
48	ABD	6	294.0	64.3	5.93	69	8.1	2.02–6.52
49	ABD	5	271.8	42.6	4.99	799	2.7	7.45–11.41

TABLE II (*continued*)

Site	Type of dyke	*N*	*D* (°)	*I* (°)	*R*	*k*	α_{95} (°)	M_0 (10^{-5} e.m.u./g)
(i) NE–SW dykes (continued)								
50	ABD	6	287.9	51.3	5.94	84	7.3	7.66–25.61
51	ABD	6	279.1	46.2	5.90	49	9.7	0.18–8.02
52	ABD	5	287.7	45.2	4.96	112	7.3	5.76–9.12
53	ABDX	5	80.6 *1	4.8	4.88	34	13.4	9.66–16.72
54	ABD	6	295.3	41.9	5.48	10	22.7	6.11–11.11
55	ABD	5	278.4	52.9	4.96	99	7.7	6.53–10.93
56	BFD	6	97.8 *1	–9.2	5.52	10	21.7	0.94–3.53
57	ABD	5	285.0	69.8	4.96	101	7.6	0.34–1.96
58	ABD	6	273.6	80.7	5.84	31	12.2	0.34–1.91
59	ABD	4	84.3 *1	67.5	3.95	66	11.4	0.05–0.42
60	ABD	5	287.1	45.4	4.84	25	15.5	0.12–1.77
61	FPD	6	266.8	53.0	5.81	26	13.4	4.31–14.87
62	ABD	6	345.2	78.3	5.97	156	5.4	1.72–6.13
(ii) BDO (WNE–ESE) dykes								
63	ABD	8	269.5	22.8	7.93	107	5.4	2.18–10.73
64	ABD	6	274.9	19.0	5.82	28	12.8	3.42–4.16
65	ABD	4	262.8	69.4	3.86	22	20.2	1.80–4.09

ABD = aphyric basalt, trachybasalt or dolerite dyke; FPD = feldspar-phyric dyke, BFD = big-feldspar dyke; TRD = trachyte dyke; X = xenocryst-bearing dyke.
*1 Dykes excluded from group mean calculation.
*2 Dyke *10* cuts dyke *11*.

Group mean directions of magnetisation and palaeomagnetic poles

	Number of sites	*D* (°)	*I* (°)	*R*	*k*	α_{95} (°)	Palaeomagnetic pole			
							lat. (°N)	long. (°E)	$d\Psi$	d_χ
(*i*)	47	289.6	50.6	44.98	22	4.5	36.4	223.9	4.1	6.1
(*ii*)	2	272.2	20.9	1.99	329	13.8	10.5	227.1	7.6	14.5

N = number of samples or sites; R = magnitude of the resultant vector with inclination I and declination D; k = precision estimate $[= (N-1)/(N-R)]$; α_{95} = semi-angle of the cone of 95%-confidence about the mean direction; $d\Psi$ and d_χ = semi-axes of the oval of confidence about the pole at the 95%-probability level in the colatitude direction and perpendicular to it, respectively.

intermediate directions may have been occupied by the palaeofield without a complete reversal being achieved. Some of the intermediate sites have comparable magnetic properties to the remaining dykes while others have lower total NRM's (Table II). In general, the intermediate sites with steep or intermediate dips have weak NRM's, while the cluster of five sites with shallow easterly directions (Fig. 7) have strong NRM's and may have been magnetised in strong transitional fields of the kind described by Shaw (1975); a definitive palaeointensity study would be needed to resolve this point further.

Thermomagnetic characteristics of typical samples from the NE–SW-trending dyke swarm are illustrated in Fig. 8. All samples have single Curie points between approximately 380 and 550°C confirming the conclusion drawn from magnetic behaviour that titanomagnetite is the remanence carrier. In most cases there is slight fall in saturation magnetisation (J_s) on cooling in air suggesting that the titanomagnetite is partially

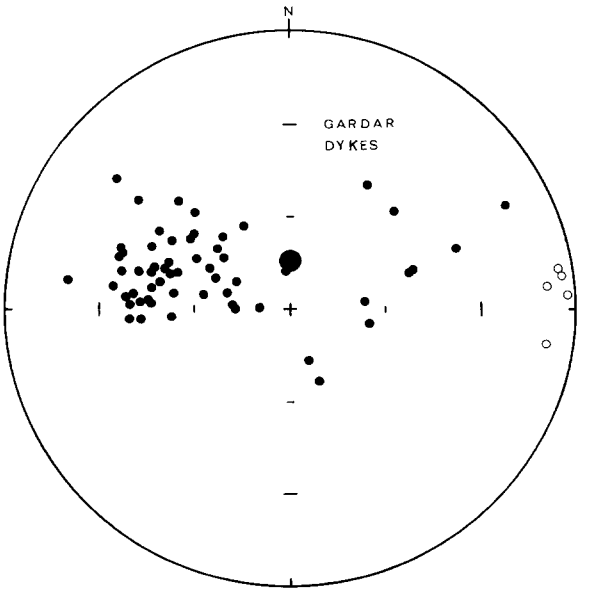


Fig. 7. Site mean directions of magnetisation for NE-trending dykes of this study.
Polar stereographic projection; closed symbols are lower-hemisphere and open symbols are upper-hemisphere plots. The large circle is the mean direction of the present dipole field in this area.

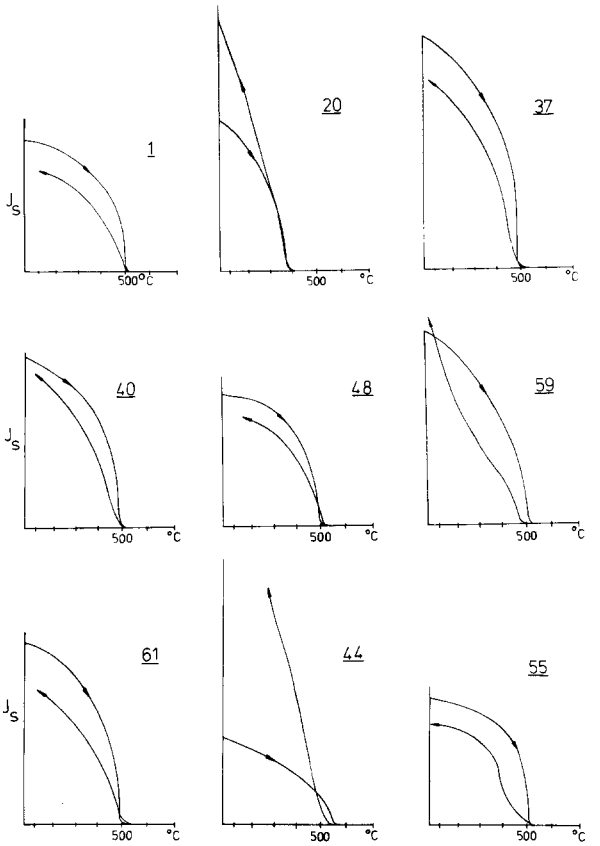


Fig. 8. Thermomagnetic curves (saturation magnetisation J_s vs. temperature) for samples from nine dykes.

oxidised to hematite (with lower J_s values) and rutile. In a few cases, however, the heating process promotes a reaction giving a marked increase in J_s on cooling (Fig. 8) and indicating production of a magnetic phase, probably after primary sulphides.

Of the three dykes belonging to the early WNW–ESE swarm, *63* and *64* have comparable declinations but much shallower inclinations than the NE-trending dykes, and combine to give a mean direction of D = 272°, I = 21° (α_{95} = 13.8°) with a palaeomagnetic pole at 227°E,10.5°N. The remaining dyke (*65*) has a direction of magnetisation remote from these two dykes, and similar to the NE-trending dykes (Table II) suggesting that it may have been remagnetised by a member of this swarm.

6. Discussion

The palaeomagnetic poles of this study from the lamprophyre dykes, dolerite dykes and Kûngnât ring dyke are moved to 189.5°E,9°N, 185°E,14.5°N and 182°E,9°N, respectively, after rotation of Greenland to North America to close the Davis Strait according to the continental reconstruction of Bullard et al. (1965). This is equivalent to an anticlockwise rotation of 22° about a Eulerian pole at 97.5°E,73°N and a clockwise rotation of 38° about a Eulerian pole at 27.7°E,88.5°N.

After this correction for Phanerozoic drift the palaeomagnetic poles are not significantly different from North American poles derived from rock units belonging to the Mackenzie igneous episode (Fig. 9a). The latter poles are derived from Mackenzie and Sudbury dykes, and Muskox intrusion, and the Coppermine Group [see Irving and Hastie (1975) for summary] and have assigned ages between 1284 and 1220 m.y. This confirms the suggestion of van Breemen and Upton (1972) that these western dyke swarms of the Gardar Igneous Province are a lateral extension of the

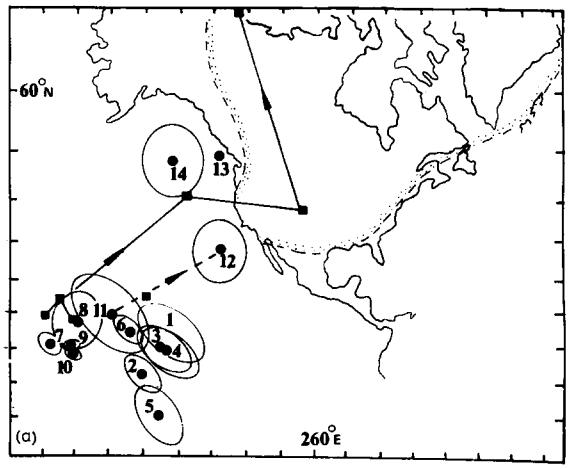


Fig. 9a. The Greenland palaeomagnetic poles plotted after closure of the Davis Strait according to the continental reconstruction of Bullard et al. (1965) with other contemporary poles from the Laurentian craton (outline stippled).
The North American poles are from: *1* = Croker Island Complex (1475 m.y.); *2* = Sherman Granite (1410 m.y.); *3* = Michikamau Anorthosite (1400 m.y.); *4* = St. Francois rocks (1375 m.y.); *5* = Sibley Group (1370 m.y.); *6* = Seal Group Redbeds (1300 m.y.); *7* = Coppermine Group (1284 m.y.); *8* = Muskox intrusion (1250 m.y.); *9* = Mackenzie diabase (1240 m.y.); *10* = Sudbury dykes (1225 m.y.); *11* = South Trap Range lavas, "reversed" group (1220 m.y.); *12* = South Trap Range lavas, "normal" group (1200 m.y.); *13* = Mamainse Point lavas, "reversed" group (1070 m.y.); *14* = Logan diabase, "reversed" sites (1160 m.y.).
North American palaeomagnetic poles are summarised by Irving and Hastie (1975) and Irving and Lapointe (1975).
Latitude and longitude are plotted at 10° intervals.

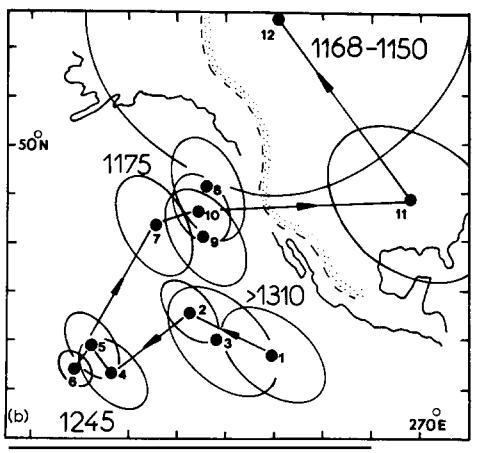


Fig. 9b. Palaeomagnetic poles from the Gardar Igneous Province, south Greenland plotted in sequence of age on present-day coordinates (latitude and longitude plotted at 10° intervals) with 95%-confidence ovals.
The numbered poles are: *1* = Gardar lavas, lower group (>1310 m.y.); *2* = Gardar lavas, upper group (>1310 m.y.); *3* = BDO dolerite dykes; *4* = NW lamprophyre dykes (<1330 m.y., >1245 m.y.); *5* = NWW dolerite dykes (<1330 m.y., >1245 m.y.); *6* = Gabbro ring dyke, Kûngnât (1245 m.y.); *7* = Hviddal Giant dyke (1175 ± 9 m.y.); *8* = Gabbro giant dykes (<1175 m.y., >1168 m.y.); *9* = Narssaq Gabbro; *10* = NE dykes (<1175 m.y., >1168 m.y.); *11* = Ilímaussaq marginal syenites (>1168 m.y.); *12* = Ilímaussaq fractionated rocks (1168 ± 21 m.y., 1150 ± 28 m.y.).
The poles plotted here which are not described in this study are listed in Piper (1977).

Mackenzie igneous episode. The Greenland rocks also have the same polarity as the Mackenzie rocks and the Gardar and Mackenzie poles are removed by 20–40° from older North American poles with assigned ages between 1300 and 1475 m.y. (Fig. 9a). This further confirms the results of Rb–Sr biotite ages (Patchett and Upton, 1977) that the dyke swarms studied here are closer in age to the Kûngnât complex (1245 m.y.) than they are to the older major intrusive episode in this area, the Grønnedal–Ika complex (Rb–Sr age 1327 ± 17 m.y.).

Upper Proterozoic palaeomagnetic data for Greenland comprise twelve palaeomagnetic poles between ca. 1400 and 1160 m.y. in age of which three are directly linked to Rb–Sr isochrons and the ages of most of the remainder are closely defined by geological relationships. The collective information is plotted in Fig. 9b and defines an apparent polar-wander (APW) movement which was extremely rapid in its later stages. The youngest of the poles is derived from highly-undersaturated syenites (Rb–Sr isochron 1168 ± 21 m.y.) in which the magnetic remance carrier is pyrrhotite, and it is in consequence poorly defined. The palaeomagnetic poles from the NE–SW dykes and the marginal facies of the Ilímaussaq intrusion are separated by 35° although they are derived from rocks intruded within a few million years of one another. The APW movement is towards Greenland and thus defines a continental movement relative to the poles of the order of 0.1 km/a; which is comparable to rates identified for certain other intervals of Precambrian time (McElhinny et al., 1974; Piper, 1976). It is possible that the distribution of magnetic inclinations in Fig. 7 reflects this APW movement, but unfortunately the youngest seg-

ment of this APW movement (poles *11* and *12*), although well dated by Rb–Sr isochrons, is based on slender palaeomagnetic evidence, and must await further results for precise clarification. However, early palaeomagnetic studies from the Keweenawan rocks of North America (Du Bois, 1962) which are in part contemporaneous with the youngest Gardar rocks, produced pole positions in NW North America and the combined evidence from the two regions supports an APW path of the form shown in Fig. 9a which has earlier been defined as the "Great Logan" palaeomagnetic loop by Robertson and Fahrig (1971).

When Greenland is restored to the pre-drift configuration of Bullard et al. (1965) the Gardar Province palaeomagnetic data move into coincidence with equivalent North American data (Fig. 9a). They firstly confirm the coherence of the continental block comprising Laurentia prior to the Grenville thermal-tectonic episode, and secondly define in more detail the direction and magnitude of contemporary polar movements. Since the youngest Gardar poles then lie in the same vicinity as poles from certain Keweenawan rocks linked to age dates between 1115 and 1070 m.y., the collective data suggest that quasi-static intervals between ca. 1280–1220 and 1150–1070 m.y. were separated by an interval of moderate to rapid APW movement.

The Gardar intrusive rocks all have the same polarity. A change in magnetic field direction from westerly positive to easterly negative is defined by Keweenawan data from the Lake Superior region (Du Bois, 1962; Robertson, 1973) and must have taken place some time after intrusion of the Ilímaussaq fractionated rocks (estimated at 1150 ± 28 m.y.) by Blaxland et al. (1977). A further reversal from easterly negative to westerly positive is well documented from study of the Mamainse Point lavas by Robertson (1973) and linked to a Rb–Sr isochron of 1070 ± 50 m.y.

Finally, the two BDO dykes which yield consistent directions of magnetisation provide a palaeomagnetic pole slightly removed from dykes with comparable trend in the western part of the province; this is moved to 211°E,15°N after correction for Phanerozoic drift and differs slightly from poles for the Mackenzie-related igneous episodes of North America. This conflicts with the Rb–Sr data but since the palaeomagnetic result is based on two sites only the difference probably has no meaning.

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